CMSC 420: Scapegoat Trees

Justin Wyss-Gallifent

January 16, 2026

1 Problems **2**

1 Problems

1. Given the following scapegoat tree:

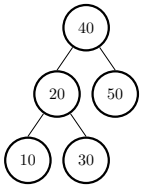

Figure 1: A Scapegoat Tree

Insert the keys $31, 32, 33, 34, 35$ in order. Restructure as needed. Explain all relevant computations. You don't need to show every intermediate tree but show enough to be clear.

2. Regarding scapegoat trees:

   (a) Suppose n is a positive integer. Prove that if n is not a power of 2 then $\lfloor \lg(n-1) \rfloor = \lfloor \lg n \rfloor$.

   (b) Use strong induction to prove the theorem from class that states that a restructured tree with n nodes will have height $\lfloor \lg n \rfloor$. Part (a) will probably be useful.

3. Suppose that at some instant we know that the height of a scapegoat tree satisfies the following with $n \geq 2$:

$$h \leq \left\lfloor \log_{3/2} n \right\rfloor + 1$$

   (a) Suppose we insert a node and do not need to rebuild the tree. Thus we now have $n+1$ nodes. Denote the new height by h'.

   Prove that the height condition is still valid. That is, prove:

$$h' \leq \left\lfloor \log_{3/2}(n+1) \right\rfloor + 1$$

   (b) Suppose we delete a node and do need to rebuild the tree. Thus we now have $n-1$ nodes. Denote the new height by h'.

   Prove that the height condition is still valid. That is, prove:

$$h' \leq \left\lfloor \log_{3/2}(n-1) \right\rfloor + 1$$

4. Suppose we insert keys $1, 2, 3, ...$ in that order into an initially empty generalized scapegoat tree with parameter α. Suppose we fix a key k and insist that α be chosen so that k will be the first inserted key which triggers a scapegoat search.

   (a) Find an inequality for α in terms of k.

   (b) Take your inequality and make it an equality and then plot this function for $1 \leq k \leq 100$ and explain why its shape makes sense. You should comment both on the general shape (increasing? decreasing?) nature of the graph as well as its behavior as $k \to \infty$.

5. Regarding Scapegoat trees, we proved a lemma in class that stated:

   If a node p has $\text{depth}(p) > \log_{3/2} n$ then p has an ancestor u for which:

$$\frac{\text{size}(u.child)}{\text{size}(u)} > \frac{2}{3}$$

   Suppose we replace $\frac{3}{2}$ by $\beta > 1$ and we want generalize the above to:

   If a node p has $\text{depth}(p) > \log_{\beta} n$ then p has an ancestor u for which:

$$\frac{\text{size}(u.child)}{\text{size}(u)} > f(\beta)$$

   (a) Following the proof from class what would be the obvious choice for $f(\beta)$? Explain.

   (b) Prove that the statement is true with this $f(\beta)$.

6. Suppose we want to reduce the frequency of rebuilding a scapegoat tree on insertion. To do this, should we increase or decrease the value of α? Explain.

7. Insert the keys $3, 2, 1, 4, 5, 6, 7, ...$ in that order into an empty scapegoat tree until the first insertion violates the scapegoat height condition. Identify the scapegoat and rebuild. You only need to show:

   - The tree when the violation occurs.
   - Justification of the violation.
   - Justification of the scapegoat.
   - The tree after the rebuild.

8. Prove that for a node u in a scapegoat tree, if u violates the scapegoat condition with $\alpha = 2/3$ and with its left child then we must have:

$$\frac{1}{2}\text{size}(u.left) > 1 + \text{size}(u.right)$$

9. This problem works through a specific example of the token calculations associated with the amortized analysis of a Scapegoat Tree. Starting with an empty Scapegoat Tree, Suppose the following sequence of operations is performed where I means insert and D means delete:

   { I:10, I:3, I:1, I:4, I:20, I:30, I:40, I:50, I:60, D:60, I:60, I:70 }

   The final insert will trigger a scapegoat hunt and rebuild but for now don't worry about that.

   (a) Draw a table showing the progress of the ID account balance during the sequence of operations.

   (b) How many tokens are in the DR account at the end? Explain.

   (c) Draw a picture of the tree and next to each node show the totals for the IR account at the end.

   (d) Find the scapegoat and explain how there will be enough tokens to cover the rebuild.

10. Suppose we insert the keys $16, 97, 81, 63, 35, 72, 55, 41, 2, 34$ into a empty scapegoat tree in that order. We then start deleting keys in the same order until the first complete restructure occurs. We then restructure and stop. What will the tree look like after the restructure?

11. Suppose we have a scapegoat tree with unknown α. Suppose we insert the keys $0, 1, 2, 3, ...$, in that order and key $k = 10$ causes a depth violation. Find the tightest possible upper bound on α.

For $k = 10$ we have $\alpha <$	

12. In the following scapegoat tree the key 60 has just been inserted and it violates the height condition. Identify the scapegoat which will be used to fix the issue (justify), and show the rebuild.

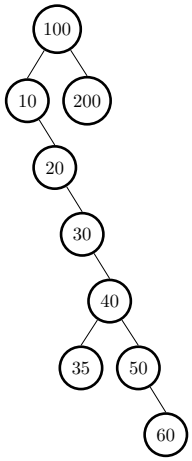

Figure 2: Scapegoat Tree Violation

13. Suppose we insert a key into a scapegoat tree with $\alpha = 2/3$ and it causes a depth violation. Is it possible for there to already be a key in the tree at the same depth? Explain.

14. Suppose we insert a key into a scapegoat tree with $\alpha = 2/3$ and it causes a depth violation. Is it possible for there to already be a key in the tree which is deeper? Explain.

15. Suppose we have a scapegoat tree with parameter $\frac{1}{2} < \alpha < 1$.

(a) As α increases does the frequency of rebuilds due to inserts increase or decrease?

Write the word INCREASE or DECREASE below.

Which?	

(b) As α increases does the frequency of rebuilds due to deletions increase or decrease?

Write the word INCREASE or DECREASE below.

Which?	

16. Suppose we have a scapegoat tree. We insert m keys and then delete keys until the instant when we have $n < \alpha \cdot m$ keys and a rebuild is triggered. For each of the following α and m, provide n. [5 pts]

α	m	n
$\frac{2}{3}$	9	
$\frac{4}{5}$	9	

17. Suppose we have a scapegoat tree with $\alpha = \frac{2}{3}$. We insert m keys and then delete keys until the instant when we have 9 keys and a rebuild is triggered. What could m have been? If you have more than one answer please list them in increasing order.

Possible m	

18. Suppose we have a scapegoat tree of height h with unknown $\frac{1}{2} < \alpha < 1$ which is perfect, having $n = 2^{h+1} - 1$ keys. Prove that inserting a key into this scapegoat tree could not possibly trigger a scapegoat-hunt.

19. Consider the following scapegoat tree:

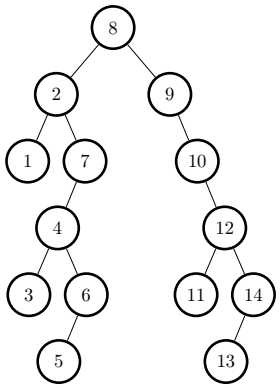

Figure 3: A Scapegoat Tree

(a) Identify the nodes which are scapegoats with $\alpha = 2/3$.

(b) For which α would there be no scapegoats at all?

20. Suppose a binary tree contains these keys:

$$4, 5, 10, 12, 13, 20, 21, 30, 50, 55$$

If we rebuild this tree, what will it look like?

21. Suppose we insert and delete nodes into a scapegoat tree according to the following pattern:

I,I,I,I,I,D,D,I,I,D,D,D,I,I,I,I,I,I,D

For this problem, ignore all rebuilding! After each operation what are the values of n and m?